

Water Quality Monitoring Report

Sampler: Lee Nulty.

Date of Sampling: 3 May 2026

Time of Sampling: 11:35 AM

Location: River Towy and River Cennen, Llandeilo, Carmarthenshire.

2. Sampling Site Coordinates

Site	Latitude/Longitude	OS Grid Ref	Context
<u>River Towy</u>	51.87810, -3.98804	SN 63248 21816	Above Iron Bridge area.
River Cennen	51.87746, -3.98910	SN 63173 21747	30m above confluence with the Towy.

1. Executive Summary

A comparative analysis of water samples taken from the River Towy and its tributary, the River Cennen, reveals distinct chemical profiles. Both rivers exhibit an alkaline pH. While the River Towy shows signs of higher nutrient enrichment (Phosphates), the River Cennen shows a concerning spike in Ammonia levels that warrants further investigation.

2. Data Comparison

Parameter	River Towy	River Cennen	Variance Analysis
pH	7.4	7.9	Cennen is significantly more alkaline; both are within normal limits.
Nitrate	1.0 mg/l	1.3 mg/l	Both rivers show low, and relatively safe levels of nitrate.
Phosphate	0.26 mg/l	0.13 mg/l	Towy is 2x higher ; indicates higher nutrient loading.
Ammonia	0.02 mg/l	0.23 mg/l	Cennen is 11.5x higher ; suggests a localized pollution event.

3. Key Findings

River Towy: Nutrient Enrichment

The Towy's Phosphate level (0.26 mg/l) is higher than the typical targets set for Special Areas of Conservation (SAC) rivers. While the water remains alkaline (pH 8.1) and low in ammonia, these phosphate levels can contribute to increased algal growth, potentially reducing oxygen levels during warmer periods.

River Cennen: Ammonia Spike

The most significant finding is the Ammonia level of 0.23 mg/l in the Cennen. With a pH of 7.9, the water is notably alkaline; in such conditions, ammonia can become more toxic to aquatic life. This reading is abnormally high compared to the Towy's near-baseline level (0.02 mg/l) and often indicates recent "fresh" contamination, such as:

- Agricultural slurry or fertiliser runoff.
- Misconnected drains or recent sewage discharge.
- Decaying organic matter near the sampling site.

4. Conclusion

The overall health of both rivers appears chemically stable in terms of pH and Nitrates. However, the Ammonia levels in the Cennen are high enough to potentially stress sensitive aquatic species, especially given the alkaline pH of 7.9.

Actions:

- 1. Upstream Check:** Survey the Cennen 500m–1km upstream of the sampling site for signs of agricultural runoff or pipe discharges.
- 2. Repeat Testing:** Re-test the Cennen within 24–48 hours to determine if the ammonia spike was a transient "pulse" or a sustained leak.
- 3. Reporting:** If the ammonia level remains above 0.20 mg/l or if visual signs of pollution (foam, grey fungus, or dead fish) are present, these results will be shared with Natural Resources Wales.